



Fruit Ripeness Monitoring System for Smart Kitchens

Group assignment for the course:
Artificial Intelligence, Machine Learning and
Deep Learning (A26_I_DTS_DL_03)

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Part 1 - The problem

Part 1.1 - Problem Definition

Distinguishing whether a fruit is in ripe or unripe stage is not always effective when done with human vision, as this might be affected by lighting, inconsistent quality standards or individual judgement. Although human inspection can be effective for a small number of fruits, it becomes more difficult to apply consistently at mass volume or over long periods of time due to tiredness, wrong standards or even time limitations.

The problem addressed in this project is the development of an image-classification model capable of recognising the visible ripeness stage of a banana from a photograph. The model analyses the visual characteristics of the banana and assigns the image to one of six categories: unripe, freshunripe, freshripe, ripe, overripe or rotten.

Part 1.2 - Objectives

The main objective of this project is to develop, train, and evaluate a multiclass image-classification system for banana-ripeness recognition.

The dataset chosen provides also an opportunity to do data exploration, learn how to preprocess images, train a classification model, evaluate its performance, and analyze the challenges of recognizing different ripeness stages.

This project can have different applications in real life and it can be extended to different kinds of fruits. The broader business objective is to reduce food waste by identifying the visible ripeness stage of fruits. This technology can be applied to the following business contexts:

- **Smart Home Kitchen Management:**

Homeowners can use the model with a smartphone app or smart appliances to monitor the ripeness of bananas and other produce in their kitchen, alerting them to consume or utilize the bananas before they become overripe, promoting healthier eating habits and reducing food waste.

- **Grocery Store Inventory Management:**

Stores can use the model to automatically monitor the ripeness of their banana stock, allowing them to more efficiently manage inventory by replacing overripe and rotten bananas while prioritizing the sale of ripe ones.

- **Produce Quality Control in Supply Chain:**
Producers and distributors can implement the model to assess the quality and ripeness of bananas during the shipping process, helping to reduce food waste by identifying and addressing ripeness issues before the produce reaches the stores.
- **Automated Crop Harvesting:**
Farmers can integrate the model into robotic harvesting systems, ensuring that only bananas at optimal ripeness stages are picked. This would streamline the harvesting process and potentially lead to higher market value for the produce, especially considering that in most cases this operation is done manually.
- **Food Industry and Recipe Recommendations:**
Recipe and meal planning apps can leverage the model to suggest tailored recipes based on the user's available banana ripeness level. For example, suggesting banana bread recipes for overripe bananas, or smoothies for ripe ones.

Part 2 - The Dataset

The **Banana Ripeness Classification dataset** used in this project is: <https://universe.roboflow.com/roboflow-universe-projects/banana-ripeness-classification>

The complete implementation of the data exploration, modelling, training, evaluation, and prediction procedures is available at the following **notebook**: https://drive.google.com/file/d/1EdA9n8_bowdmPuppNUoLiwfMIpKyz0pB/view?usp=drive_link

Part 2.1 - The Dataset & Classification Task

The dataset contains 5,616 labelled images of bananas photographed at different visible ripeness stages. The images are divided into three subsets: training images, validation images, and test images. The dataset contains six classes: *unripe*, *freshunripe*, *freshripe*, *ripe*, *overripe* and *rotten*. The labels are inferred from the names of the folders containing the images.

Dataset structure within './Banana_Ripeness_Dataset':

```
Banana_Ripeness_Dataset/  
  valid/  
    overripe/  
    ripe/  
    rotten/  
    freshripe/  
    unripe/  
    freshunripe/  
  test/  
    freshripe/  
    freshunripe/  
    overripe/  
    ripe/  
    rotten/  
    unripe/  
  train/  
    ripe/  
    rotten/  
    freshunripe/  
    unripe/  
    overripe/  
    freshripe/
```

The model should learn to identify the visible ripeness stage of a banana based on visual characteristics such as peel colour, brown spots, texture, and signs of deterioration. This is a multiclass classification problem, each image belongs to one of six mutually exclusive categories.

The team chose this dataset because it addresses a relevant problem connected to food-waste reduction and provides a clear, realistic image-classification task. Finally, the presence of visually similar categories such as freshripe/ripe and unripe/freshunripe creates a meaningful challenge for analysing classification limitations and model generalization.

Part 2.2 - Training/Validation/Test Sets

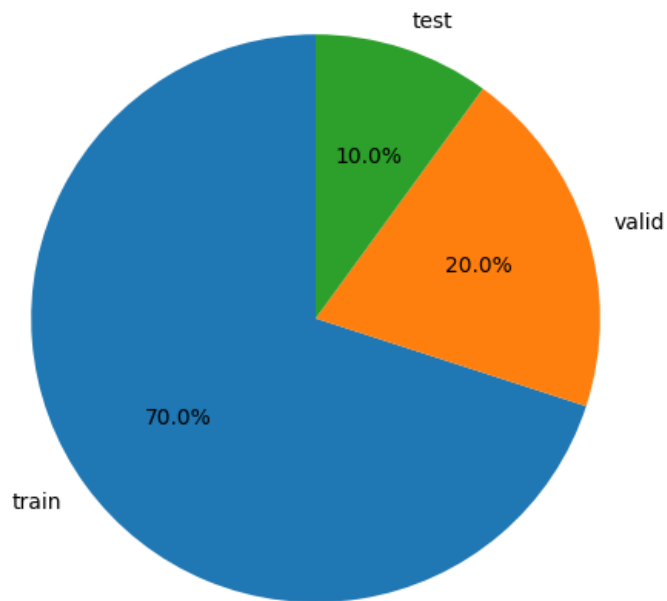
The images in the dataset are organised into three separate subsets:

- a *training set* used to learn the visual patterns associated with each class;
- a *validation set* used to monitor the model during training;
- a *test set* used for the final evaluation.

The dataset contains 5,616 images in total, divided as follows:

Dataset Split	Number of images	Percentage
Training Set	3913	70%
Validation Set	1123	20%
Test Set	562	10%

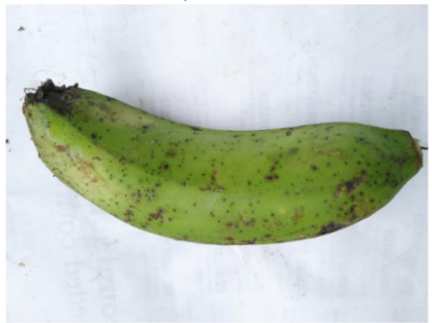
Distribution of Images Across Subsets

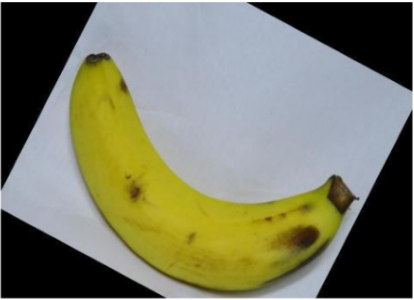


Part 3 - The Classes

Part 3.1 - Class Description

The dataset contains six classes representing different banana ripeness conditions:

Class	Description	Representative image
Unripe	These bananas are intensely green usually with dark spots, similar in color to freshunripe class, however, they are less developed and require more time to ripen fully.	Unripe (train set) 

Freshunripe	Bananas in this category are predominantly green in color, indicating they are not yet ripe. These bananas are often chosen for cooking.	Freshunripe (train set) 
Freshripe	These bananas typically have a bright yellow peel, often with some green remaining at the tips. There are usually no brown spots or blemishes, indicating peak freshness and ripeness for consumption.	Freshripe (train set) 
Ripe	Ripe bananas have a vibrant yellow peel, usually with a few scattered brown speckles. This is generally considered the optimal stage for eating raw, offering a good balance of flavor and texture.	Ripe (train set) 
Overripe	Overripe bananas are characterized by a yellow peel extensively covered with brown or black spots, or even entirely black. While not ideal for eating raw, they are excellent for baking due to their high sugar content and soft texture.	Overripe (train set) 

Rotten	Rotten bananas exhibit significant decay. Their peels are typically completely black, often with signs of mold or decomposition. These are usually past the point of consumption and should be discarded.	Rotten (train set) 
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The images consist of a single, centered banana captured from varying angles, backgrounds vary across wood, white paper, tile, and colored cloth. This introduces potential risk for spurious correlation if specific classes correlate with specific backdrops. Furthermore the ripening is a continuous biological process discretized into hard categories:

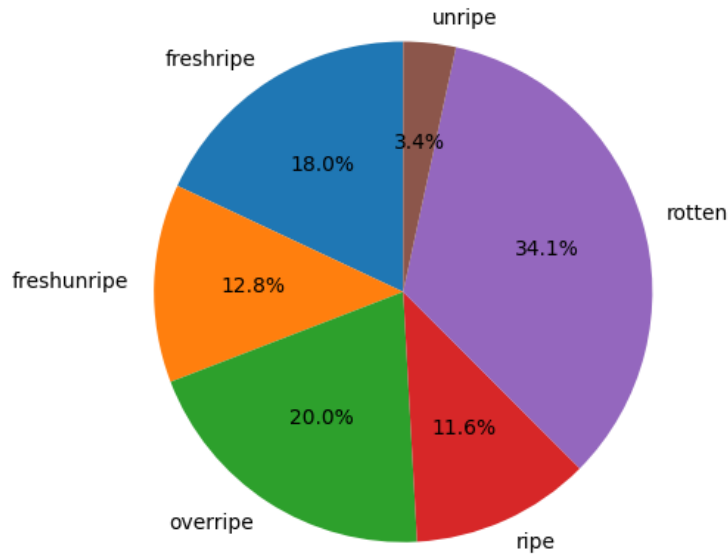
- unripe vs freshunripe: green skin, separated primarily by spot density.
- freshripe vs ripe: yellow skin, separated by subtle dark streaking and spotting.
- ripe vs overripe vs rotten: continuous increase in black/brown necrotic coverage rather than distinct categorical features, leading to label noise at adjacent boundaries.

Part 3.2 - Class Distribution

The dataset breakdown in classes is the following:

Class	Number of images	Percentage
Unripe	191	3.4%
Freshunripe	720	12.8%
Freshripe	1,013	18.0%
Ripe	654	11.6%
Overripe	1,125	20.0%
Rotten	1,913	34.1%

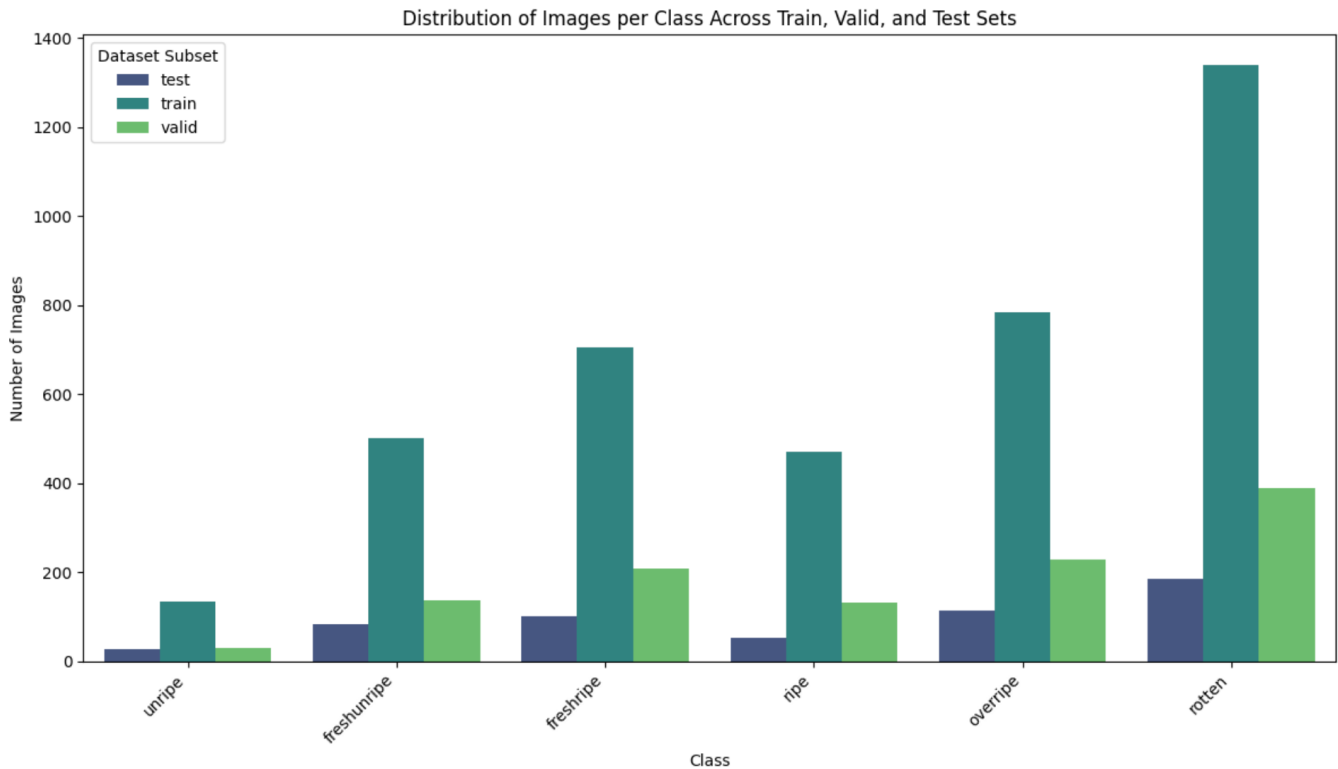
Distribution of Images Across All Classes



The rotten class represents the single largest category, accounting for 34.1% of the overall dataset. Combined with the overripe class (20.0%), non-fresh or degraded states comprise over 54% of the entire collection. This provides the model with strong supervision for identifying defective or post-shelf-life produce. There is a pronounced visual imbalance between the majority class (rotten at 34.1%) and the minority class (unripe at 3.4%), a ratio of roughly 10:1. Without class weighting or focal loss, standard classifiers bias heavily toward majority classes, so that overall accuracy should not be interpreted on its own but precision, recall, F1-score and the confusion matrix should be also considered to obtain a more complete evaluation.

Based on the image counts and percentage distributions across the training, validation, and test datasets, the following class imbalance observations have been identified:

Set Split	Training		Validation		Test	
Class	Count	Share	Count	Share	Count	Share
Unripe	133	3.4%	31	2.8%	27	4.8%
Freshunripe	501	12.7%	136	12.1%	83	14.8%
Freshripe	704	17.9%	207	18.4%	102	18.1%
Ripe	470	12.0%	132	11.8%	52	9.3%
Overripe	783	19.9%	229	20.4%	113	20.1%
Rotten	1340	34.1%	388	34.5%	185	32.9%



The *training set* shows 'rotten' as the most prevalent class, being approximately 10 times larger than the 'unripe' class. This imbalance could lead to models biased towards classifying images as 'rotten' and potentially underperforming on 'unripe' images.

The *validation set* mirrors the training set's imbalance, with 'rotten' remaining the dominant class and 'unripe' being critically underrepresented. This consistent distribution helps ensure that the validation performance is a realistic indicator of the model's behavior on similar imbalanced data.

The *test set* also demonstrates a similar class distribution, though 'unripe' shows a slightly higher percentage compared to the training and validation sets, it remains significantly smaller than 'rotten'. Evaluating this set will confirm how well the model generalizes to unseen data with comparable class frequencies.

The observed severe class imbalance, particularly the disproportionately high number of 'rotten' images and very low number of 'unripe' images, suggests that a standard training approach might result in a model that performs poorly on minority classes. Strategies such as data augmentation (especially for minority classes), oversampling minority classes, undersampling majority classes, using weighted loss functions, should be considered during model development to mitigate this issue and improve generalization across all ripeness stages.

Part 4 - Data Exploration

Data exploration was performed before model training in order to understand the visual content, quality, structure, and potential limitations of the dataset. The analysis focused on image examples, class-specific visual patterns, image resolution, colour information, and the suitability of the data for image classification.

Part 4.1 - Image Examples

The images are similar at the semantic level because they all show bananas, as single fruits and approximately centred in the frame. However, they vary in visual appearance and photographic conditions (e.g. lighting and brightness, camera angle, background).

The banana shape and framing are relatively consistent across the dataset. This is useful because the model can focus mainly on the visual characteristics related to ripeness rather than having to identify completely different objects.

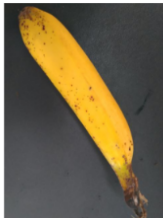
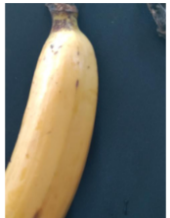
The main differences between the classes can be summarised as follows:

- *unripe* and *freshunripe*: predominantly green skin; *freshunripe* is clean and smooth, while *unripe* may show brown spotting and bruising on green skin.
- *freshripe* and *ripe*: predominantly yellow skin; *freshripe* is smoother pale yellow, while *ripe* may show more dark streaking and spotting.
- *overripe*: yellow-brown with heavy black spotting/streaking.
- *rotten*: extensive black/brown necrotic patches, sometimes on skin that is still partly green.

Backgrounds vary considerably (white paper, tile, wood, colored cloth), which is worth checking for spurious correlation with class, if certain classes were systematically photographed on a particular backdrop, the model could shortcut on background rather than the banana itself.

The following figures, extracted from the dataset, show that the classes follow a general visual progression from green to yellow, brown, and black. However, the boundaries between the classes are not always sharp.



Freshunripe					
Freshripe					
Ripe					
Overripe					
Rotten					

The classification task for a human is relatively easy for clearly green bananas or visibly deteriorated bananas, but moderately difficult for intermediate ripeness stages.

The main pairwise comparisons are:

- *unripe* versus *freshunripe*: generally easier to distinguish because of differences in cleanliness, spots, and bruising on the green peel.
- *freshripe* versus *ripe*: usually distinguishable through colour intensity and spot density, although borderline examples exist.
- *ripe* versus *overripe*: more difficult because both classes can contain yellow-brown peel and dark spots (several images from each class are close to indistinguishable, the boundary looks like a browning-percentage threshold rather than a hard visual line).
- *overripe* versus *rotten*: often ambiguous because the distinction may depend on the extent and severity of dark or necrotic areas.

Banana ripening is a continuous process, but the dataset represents it using six discrete classes. Label noise near adjacent-class can be expected, and also misclassification from the model (ripe $\leftrightarrow$ overripe, overripe $\leftrightarrow$ rotten). Instead between visually distant classes (e.g., unripe vs. rotten) should be easier.

There could be ambiguity between the classes. Here a few examples:

- a banana with mostly a yellow peel with a few brown spots $\rightarrow$ difficult to distinguish between the ripe and overripe classes

Ripe (Train set)	Overripe (Train set)
	

- Images with high brightness could also make classification more difficult

Freshunripe (Train set)	Freshripe (Train set)

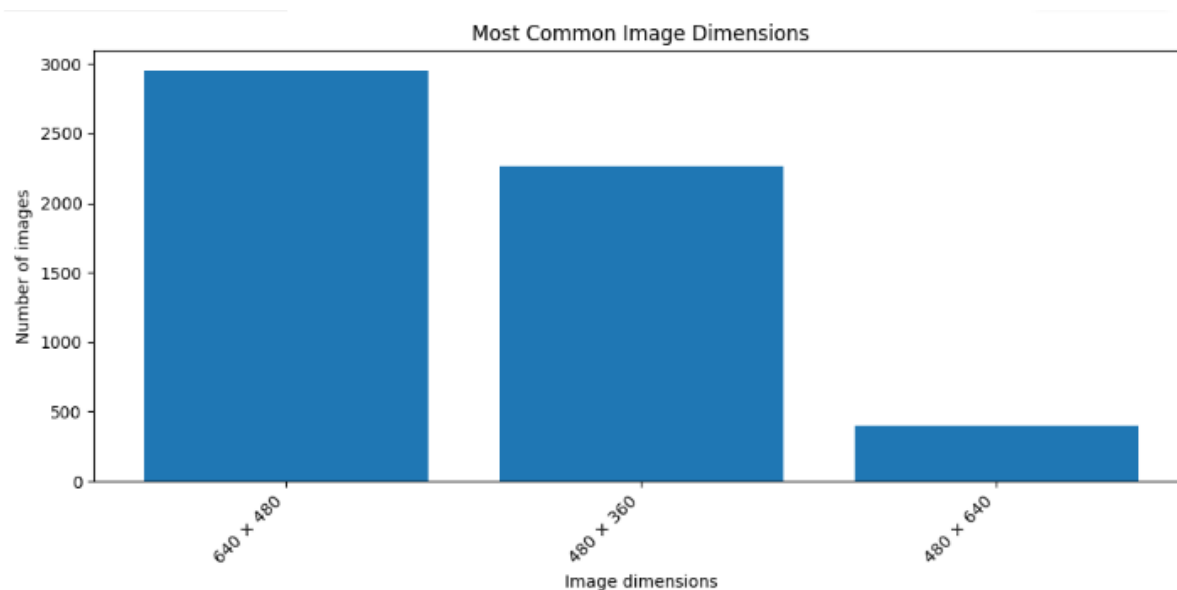
These ambiguous cases may lead to confusion between classes and reduce the model's accuracy.

Instead, the model could distinguish between classes based on those characteristics:

- **Overall hue:** the green → yellow → brown/black progression is the primary ripeness signal.
- **Spot/blemish density and coverage:** a simple dark pixel fraction proxy roughly doubles from unripe/freshunripe to overripe/rotten, and separates fresh vs. non-fresh and ripe vs. overripe.
- **Texture:** smooth, taut skin vs. wrinkled/collapsing skin as rot progresses.
- **Spot pattern and shape:** small speckles vs. large confluent black patches (rotten).

Part 4.2 - Data Quality

The original images were all stored in .jpeg format and using RGB color mode, but using different resolutions. The most frequent resolutions were:



Resolution	Number of images
640 x 480	2,948
480 x 360	2,266
480 x 640	402

The dataset includes both landscape and portrait images.

Part 4.3 - Color Analysis (RGB and HSV)

Colour analysis was performed to investigate whether simple global colour statistics could provide useful information for distinguishing banana-ripeness stages.

For each image, six colour features were extracted:

- mean Red
- mean Green
- mean Blue
- mean Hue
- mean Saturation
- mean Value.

The RGB images were converted to floating-point arrays and normalised to the range [0,1]. Then, the images were then converted from RGB to HSV using OpenCV. Hue was normalised from the OpenCV range 0°–360° to [0,1], while Saturation and Value were already represented in the range [0,1].

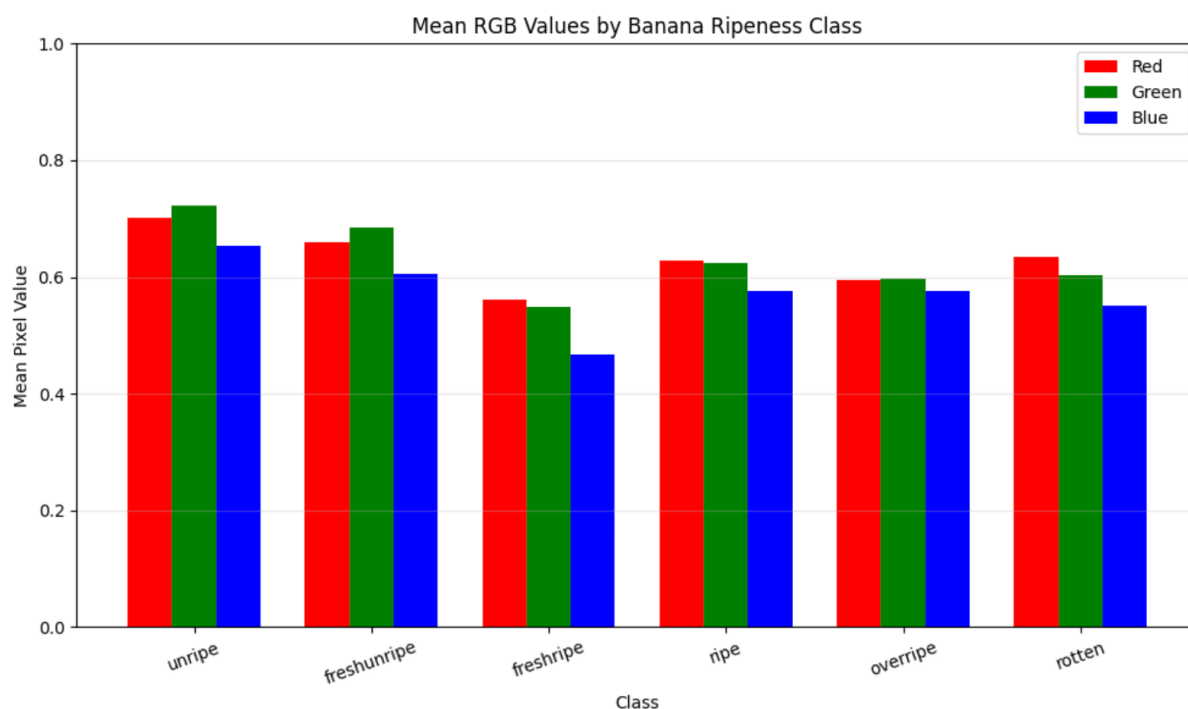
The purpose of this analysis was not to replace the CNN model, but to understand how much information could be obtained from simple colour statistics. The extracted features were also used as inputs for a Logistic Regression and Random Forest models.

The following table reports the average colour statistics for the test set.

Class	Mean R	Mean G	Mean B	Mean H	Mean S	Mean V
Unripe	0.717	0.737	0.661	0.452	0.206	0.753
Freshunripe	0.663	0.690	0.607	0.420	0.223	0.708
Freshripe	0.565	0.556	0.466	0.422	0.300	0.610
Ripe	0.640	0.631	0.587	0.440	0.264	0.693
Overripe	0.597	0.598	0.577	0.431	0.181	0.633
Rotten	0.640	0.610	0.562	0.414	0.210	0.753

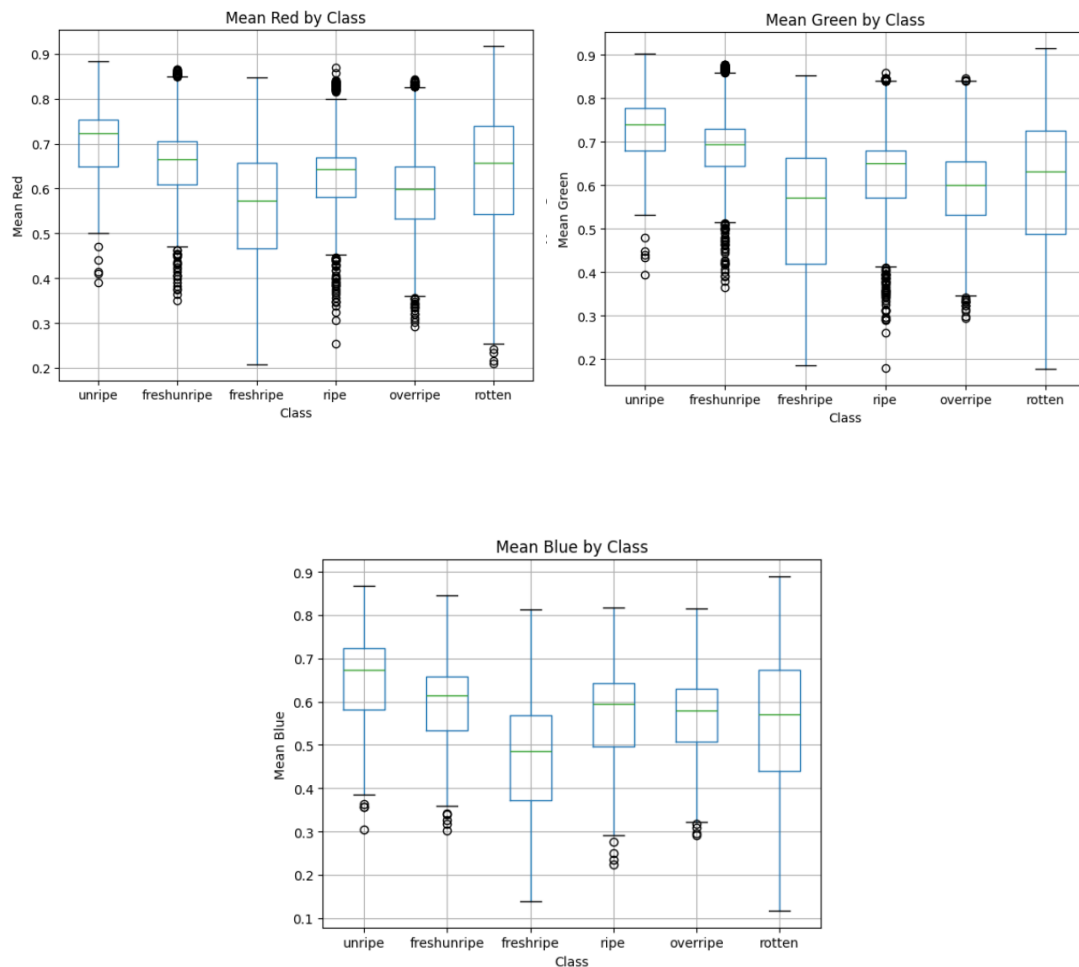
The results indicate that some colour statistics differ across classes. For example, unripe has the highest average RGB values and the highest Value, while freshripe has the highest Saturation. Overripe has the lowest average Saturation, which may be related to its more brownish and less vivid appearance.

However, the differences are not sufficient to clearly separate all classes. In particular, several neighbouring categories have similar global colour values. Ripe and rotten show similar average RGB values, while freshunripe and unripe have similar Hue values. This overlap suggests that mean colour statistics alone cannot completely represent the visual complexity of banana ripeness.



Analysing the Mean RGB values, the classes that overlaps the most are ripe/overripe and unripe/freshunripe. For these classes it is expected the model to make some misclassification because the mean values of RGB are similar. Meanwhile for freshripe and rotten the average values seem to be more distinct leading to more precise classification outcomes.

Separately analyzing the Mean RGB, it is possible to observe a more precise distinction between each classes, especially within unripe and freshripe classes (-+ 0.2 points). On the other hand, ripe and overripe class have still low mean variation (-+ 0.1 point).



The HSV plots show that Value and Saturation provide some degree of class separation, but the distributions overlap considerably.

The following patterns can be observed:

- unripe generally has higher Value values
- freshunripe also tends to have relatively high brightness
- freshripe shows comparatively high Saturation
- overripe has lower Saturation values
- ripe and overripe overlap, reflecting their similar visual appearance
- overripe and rotten also overlap because both may contain brown or dark areas.

An analysis of the box plots of mean HSV values shows no distinct differences within the mean values of Hue for each class. However, there is a small but visible difference in 0.05 point within mean Saturation and Value means within the classes.

The box plots are therefore useful for describing the dataset, but they do not prove that one feature can independently classify the images. The distributions

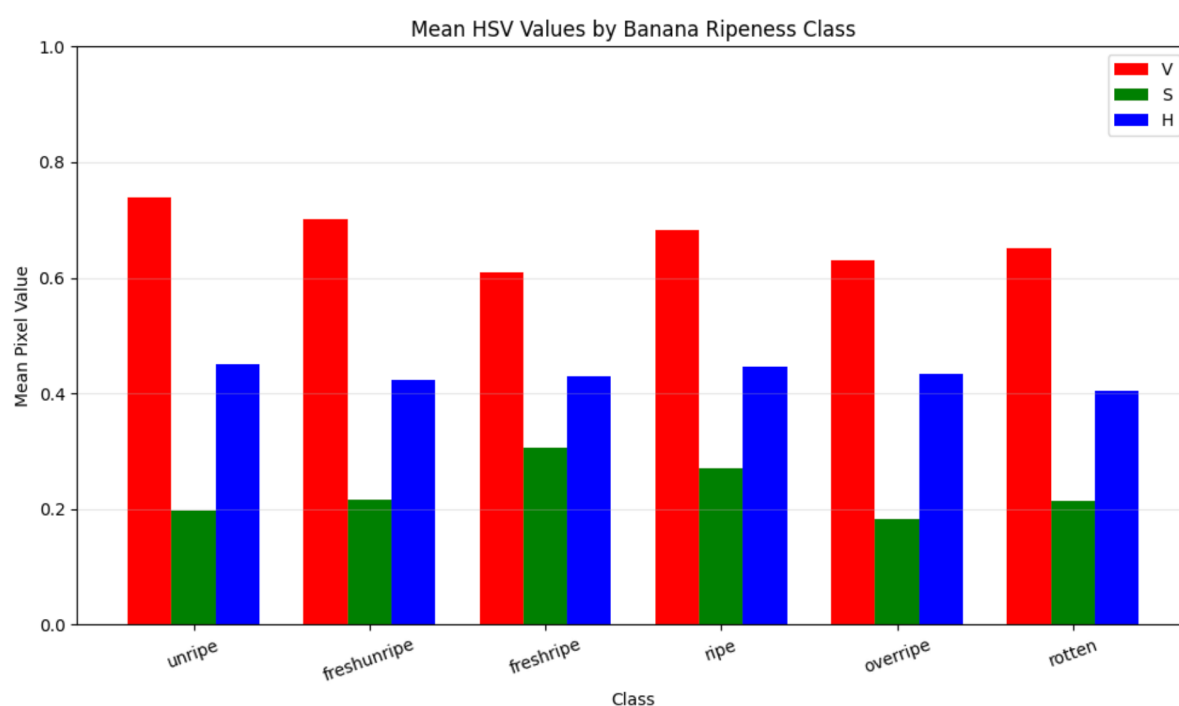
should be interpreted as exploratory evidence rather than as definitive class boundaries.

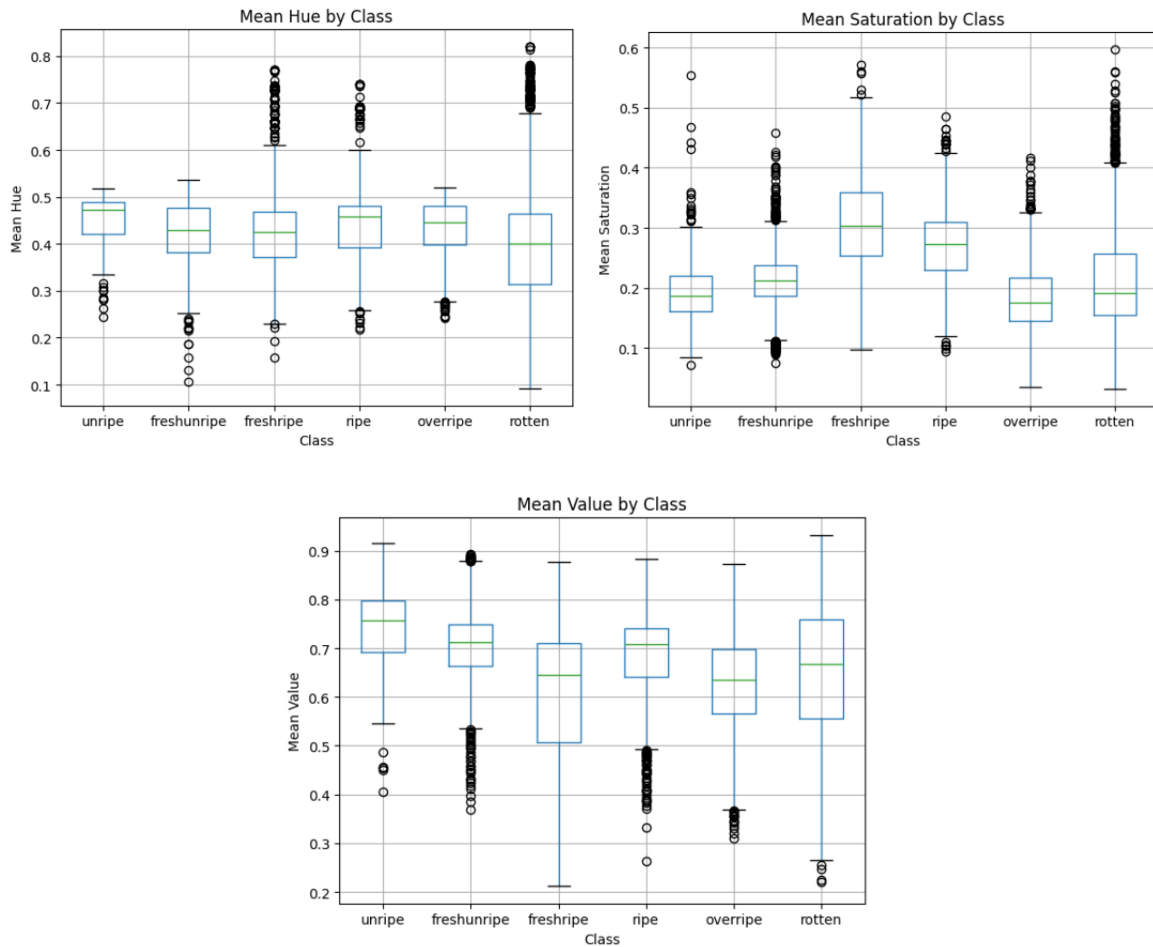
The extracted features are global averages calculated over the entire image. They describe the overall colour composition but do not preserve spatial information. For example, two images may have similar average colour values even though one contains a mostly yellow banana with a small dark damaged area, while the other contains a banana with a uniformly brownish peel.

The average RGB and HSV values could be similar, but the local visual patterns are different. Global features do not indicate:

- where the dark spots are located
- how large the spots are
- whether the banana includes different colour regions.

This limitation is important because ripeness and deterioration often depend on local patterns rather than only on the average colour of the whole image.





Part 5 - The Model

Part 5.1 - Machine Learning and Deep Learning Model

This project compares traditional machine learning and deep learning for banana-ripeness classification. The machine-learning approach uses manually extracted global RGB and HSV colour features as input to Logistic Regression and Random Forest models. The deep-learning approach uses a convolutional neural network that learns visual features directly from the images.

Due to potential cost constraints of computational resources, it is good practice to approach a dataset with an incremental technique. Traditional machine-learning models were first used as baselines because they are relatively simple, computationally efficient, and easier to interpret, then adding complexity to reach the best potential accuracy for our image classifier. The results were then compared with a CNN-based model.

The comparison evaluates whether global colour statistics are sufficient for the classification task or whether automatically learned spatial features—such as edges, spots, textures, and local colour patterns—provide better performance.

Part 5.2 - Logistic Regression

Logistic Regression was used as the first machine-learning baseline for the banana-ripeness classification problem. It provides a simple and interpretable reference model before moving to more complex machine-learning and deep-learning architectures.

Logistic Regression is a classification algorithm that models a relationship between predictor variables and a categorical response variable. The Logistic Regression could be of three types: Binary, Nominal and Ordinal. In this study it is employed a multinomial logistic regression since the categorical response variable is not a binary classification problem but a multi class problem.

The predictor variables are the mean values of RGB and HSV, the input of the model. The categorical response variable is the ripeness and the state of decay of banana, represented by the six classes. This model estimates the probability that an observation belongs to a specific response variable.

The model therefore follows this structure:

Input images

→ **Mean RGB and HSV feature extraction**

→ **Multinomial Logistic Regression**

Part 5.3 - Random Forest

Random Forest was used as a second machine-learning baseline. Random Forest is an ensemble machine learning algorithm that combines multiple Decision Trees to perform classification or regression tasks. Each tree is trained on a different bootstrap sample of the training data and considers a random subset of features when determining the splits. The final prediction is obtained by aggregating the predictions of the individual trees, for example through majority voting in classification problems.

Unlike Logistic Regression, which models the relationship between the input features and the target variable using a linear decision boundary, Random Forest can capture non-linear relationships and interactions between features without requiring these relationships to be explicitly defined.

In this project, Random Forest was used as a complementary model to Logistic Regression to investigate whether the relationship between the image-derived features and the banana ripeness classes could be better described by non-linear patterns. The input features were extracted from the images and included colour-related measurements such as the mean RGB and HSV values. Since different stages of banana ripeness may be associated with complex combinations of colour characteristics, Random Forest provides a way to capture potential interactions and non-linear patterns between these features and the target classes.

The model follows this structure:

Input images

- **Mean RGB and HSV feature extraction**
- **Multiple Decision Trees**

Part 5.4 - CNN - MobileNetV2

We used MobileNetV2 as the base of the classifier. MobileNetV2 is a relatively lightweight convolutional neural network that can be used with transfer learning. Instead of training all visual features from the beginning, we reused weights learned from ImageNet and adapted the final classification stage to the banana-ripeness task. This approach was selected because it reduces training time and computational requirements while providing a strong baseline for a dataset of moderate size. The original ImageNet classification head was removed because it predicts ImageNet categories rather than banana-ripeness categories. The MobileNetV2 base was frozen, meaning that its pre-trained weights were not updated during the training stage. Only the new classification head was trained. This strategy reduces the risk of changing useful pre-trained visual features too quickly and makes the model easier to interpret.

The model architecture is:

Input images [dimensions: 224x224x3]

- **MobileNetV2** [pretrained base, frozen weight]
- **GlobalAveragePooling2D**
- **Dropout** [parameter 0.2]
- **Dense** [6 neurons, softmax activation]

Each image was loaded in RGB format and resized to 224 × 224 pixels. Therefore, the model input has the shape (None, 224, 224, 3), where the first

dimension indicates that the batch size is variable, instead the final dimension represents the red, green, and blue colour channels.

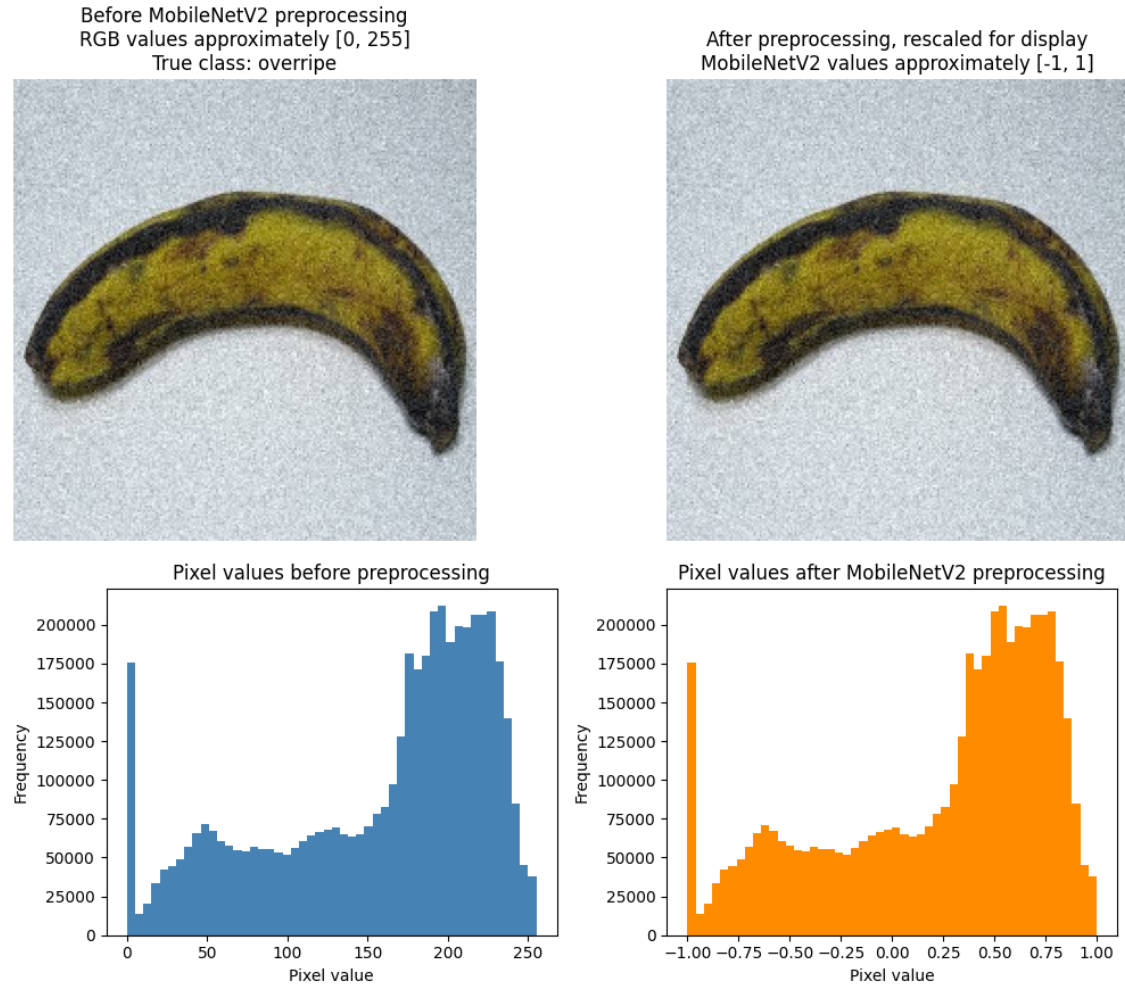
The MobileNetV2 preprocessing function was included in the model. It rescales the original pixel values from approximately $[0, 255]$ to the range $[-1, 1]$, which is the range expected by the pre-trained MobileNetV2 weights.

The output of MobileNetV2 is a set of convolutional feature maps with shape (None, 7, 7, 1280).

GlobalAveragePooling2D converts these feature maps into a (None, 1280) element feature vector by averaging each feature spatially. This creates a compact representation, it is also a necessary layer, because the dense layer needs a 2D vector as input.

A Dropout layer with a rate of 0.20 (20% of neurons are dropped during training) was added to reduce the risk of overfitting in the newly trained classification head. The final Dense layer contains six output neurons, one for each ripeness class. The softmax activation converts the output scores into probabilities that sum to one. The class with the highest probability is selected as the model prediction.

The final model contained 2,265,670 parameters. Of these, 7,686 were trainable parameters in the final classification layer, while 2,257,984 parameters belonged to the frozen MobileNetV2 base.



Part 6 - Training & Performance

To evaluate the performance of the classification model, several metrics were calculated based on the logistic regression and the random forest results obtained on the test set. These metrics provide a more detailed assessment of the model's predictive performance, including accuracy, precision & recall, F1-score, ROC & AUC and the confusion matrix.

Part 6.1 - Accuracy

$$\text{Accuracy} = \frac{\text{correct classifications}}{\text{total classifications}} = \frac{TP + TN}{TP + TN + FP + FN}$$

Formula stating how Accuracy is obtained.

Logistic Regression:

In this phase due to fast evaluation between models, the focus is centered on the test classification report where accuracy, precision and recall are evaluated.

The accuracy is given thanks to the ratio of correct classifications over total classifications, how many true positives and true negatives are detected by the model over all the classifications. A value of 0.52 indicates that the model makes 52% of correct classification and 48% wrong classifications, as an approximation the model makes 5 correct predictions out of 10.

Metric	Result
Validation Accuracy	53%
Test Accuracy	52%

Random Forest:

Random Forest model improved by 55% the accuracy of the prediction with an overall value of 0.81 with respect to the Logistic Regression. Even though it has to be considered if the model overfitted the dataset or it just outperformed well on the most representative rotten class.

Metric	Result
Validation Accuracy	83%
Test Accuracy	81%

Part 6.2 - Precision & Recall

Going deeper through the test classification report, the precision, that states the ratio between correctly classified actual positives over both correct positive and false positives, indicates for each prediction made, how much these are trustworthy, and it shows that ripe was the class with high overall meaning that there is a low percentage of misclassified examples.

$$\text{Precision} = \frac{\text{correctly classified actual positives}}{\text{everything classified as positive}} = \frac{TP}{TP + FP}$$

$$\text{Recall (or TPR)} = \frac{\text{correctly classified actual positives}}{\text{all actual positives}} = \frac{TP}{TP + FN}$$

Logistic Regression:

The precision values vary considerably across the different classes. The highest precision is obtained for the ripe class (0.71), meaning that approximately 71% of the samples predicted as ripe are correctly classified. In contrast, the freshunripe class has a much lower precision (0.17), while the unripe class has a precision of 0.00, due to the few number of images in this class.

The Recall, that states the ratio between correctly classified actual positives over all actual positives indicates if the model is able to recognize all the samples belonging to a class, a high percentage related to a class rotten means that the model can easily recognize it correctly, with a low chance of false negative.

Class	Precision	Recall	F1-score	Support
Unripe	0%	0%	0%	27
Freshunripe	17%	1%	2%	83
Freshripe	54%	62%	58%	102
Ripe	71%	19%	30%	52
Overripe	55%	58%	56%	113
Rotten	50%	83%	62%	185

Random Forest:

The parameters of Precision and Recall for the Random Forest are definitely better, also in this case.

Except for the ripe class, the values of Precision are higher than 0.8 (0.8%), meaning that we have few cases of false positive.

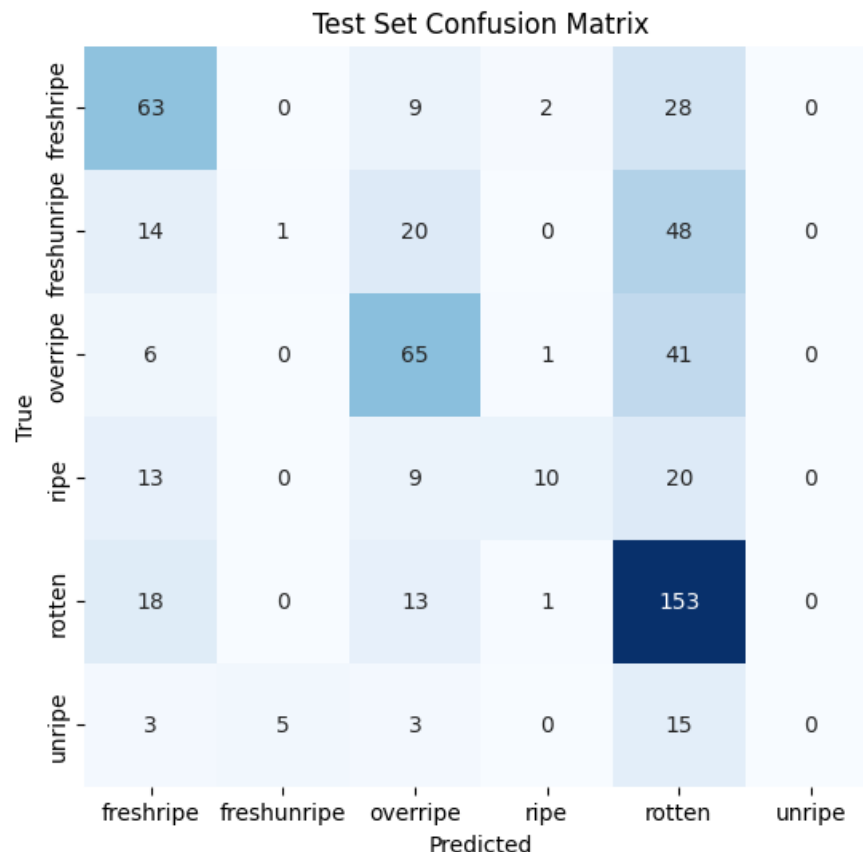
The Recall is in general higher than the Logistic Regression, with an average value of 0.8 (80%).

Class	Precision	Recall	F1-score	Support
Unripe	92%	46%	62%	27
Freshunripe	87%	86%	86%	83
Freshripe	80%	83%	82%	102
Ripe	74%	62%	67%	52

Overripe	80%	88%	84%	113
Rotten	82%	85%	84%	185

Part 6.3 - Confusion Matrix

Logistic Regression:



The confusion matrix put on the x-axis the predicted classes and on the y-axis the ground truth. It shows looking at the degree of the color, the more the color is darker, the more the predicted values correspond to the ground truth if the coordinates are the same, i.e. the rotten class is the one with the high correct values obtained with low errors (153 images were correctly classified as rotten and they were rotten, meanwhile 18 images were classified as freshripe when they were in reality rotten).

An ideal confusion matrix should have the diagonal with high concentration meaning that each predicted number of classes belong to the actual classes. Unripe class was completely misclassified, since we have 0 on the points (unripe, unripe) alike freshunripe with just 1 image classified correctly.

Random Forest:

Random Forest Test Set Confusion Matrix

True	freshripe	freshunripe	overripe	ripe	rotten	unripe
	85	2	3	2	10	0
	4	71	3	1	4	0
	3	0	100	6	4	0
	8	1	4	32	7	0
	6	3	15	2	158	1
	freshripe	freshunripe	overripe	ripe	rotten	unripe

The Random Forest, based on the confusion matrix above, performs well overall, showing 82% of accuracy, with high results for overripe and rotten classes. The main weakness appears in ripe and unripe classes, which get frequently confused. This problem is due to class imbalance and overlapping visual features between adjacent ripeness stages.

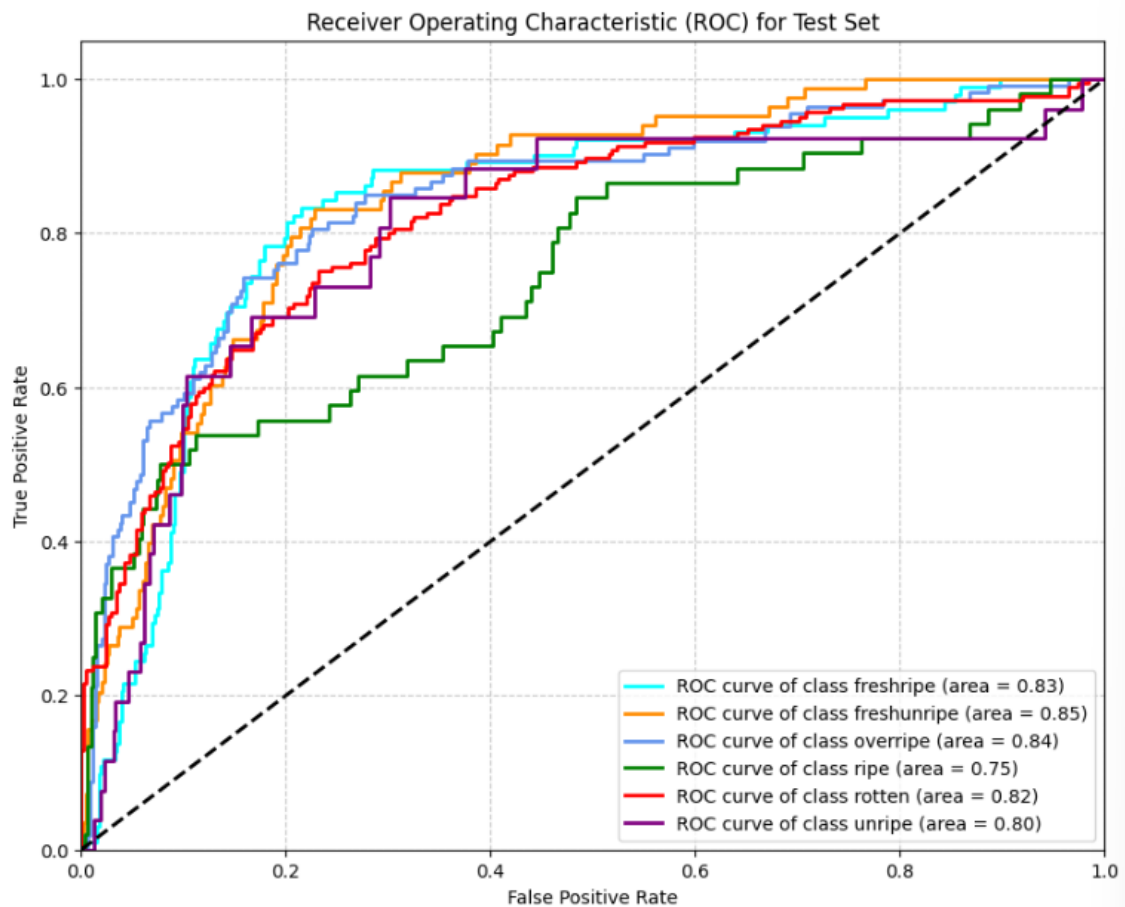
Part 6.4 - ROC & AUC

In this project we used the ROC (Receiver Operating Characteristic) and the AUC (Area Under the Curve) as additional metrics to evaluate the results obtained from our model.

The ROC curve evaluates the ability of the model to discriminate between each target class and the remaining classes by plotting the True Positive Rate against the False Positive Rate across different classification thresholds.

The AUC provides a single numerical summary of this discriminative ability, with values ranging from 0.5, corresponding to random discrimination, to 1.0, corresponding to perfect discrimination. Higher AUC values indicate a greater ability to distinguish the target class from the other classes.

Logistic Regression:



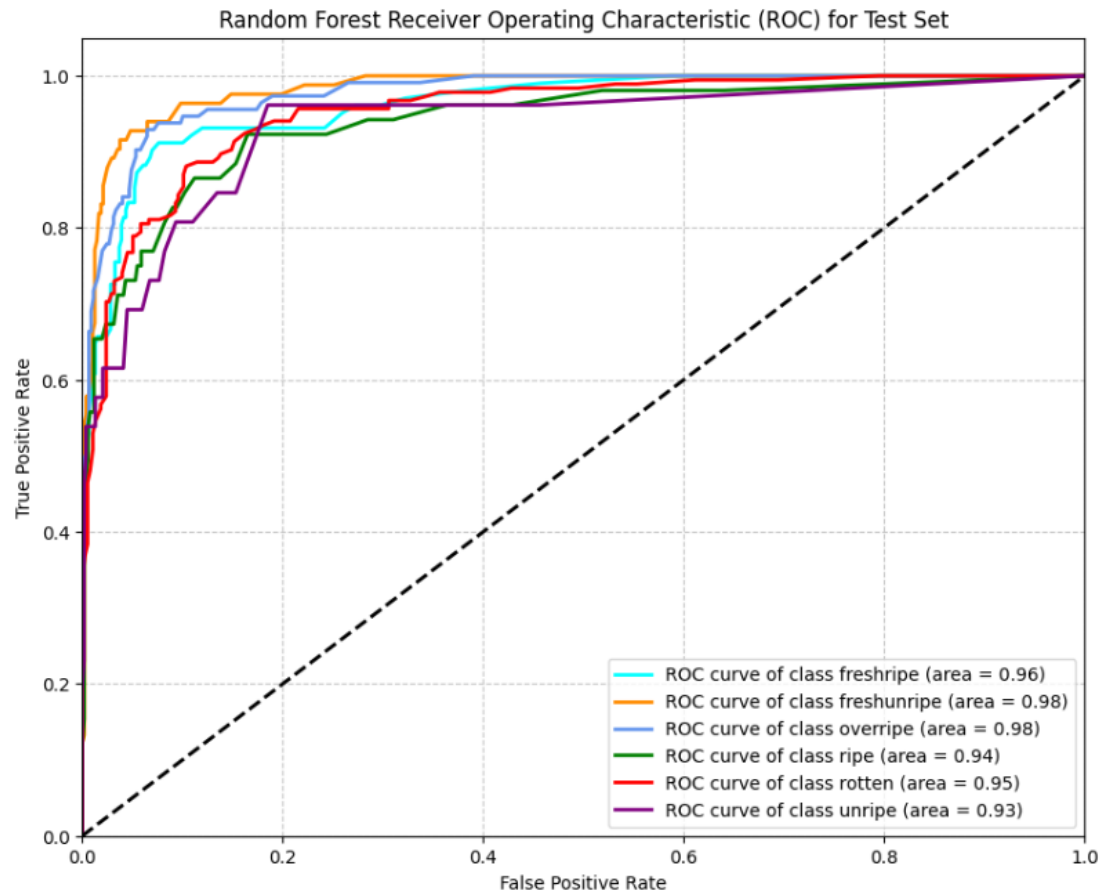
Class	AUC
Unripe	0.8
Freshunripe	0.85
Freshripe	0.83
Ripe	0.75
Overripe	0.84
Rotten	0.82

The **freshunripe class has the highest AUC (0.85)**, meaning that the model is relatively effective at distinguishing fresh bananas from the other classes.

The **ripe class has the lowest AUC (0.75)**, indicating that this class is more difficult to distinguish. This could be because ripeness is a gradual process,

meaning that visually similar characteristics may occur between neighbouring stages or that due to low samples from this class, the model had difficulty in detecting a distinct pattern.

Random Forest:



Class	AUC
Unripe	0.93
Freshunripe	0.98
Freshripe	0.96
Ripe	0.94
Overripe	0.98
Rotten	0.95

The **freshunripe and overripe class has the highest AUC (0.98)**, meaning that the model separates these classes from the rest of the dataset almost perfectly.

The **unripe class has the lowest AUC (0.93)**, followed by **ripe (0.94)**. This info is consistent with the confusion matrix, confirming these two classes are the hardest for the model to distinguish.

Part 6.5 - Training and Evaluation of CNN

The dataset was divided into training, validation, and test sets. The training set was used to update the trainable parameters. The validation set was used during training to monitor generalization and identify possible overfitting. The independent test set was reserved for the final evaluation.

Images were loaded in batches of 32. The model was compiled with the Adam optimiser, a learning rate of 0.001, sparse categorical cross-entropy loss, and accuracy as the main training metric. Sparse categorical cross-entropy was appropriate because the labels were represented as integer class identifiers.

The maximum number of training epochs was set to 20 and two callbacks were used:

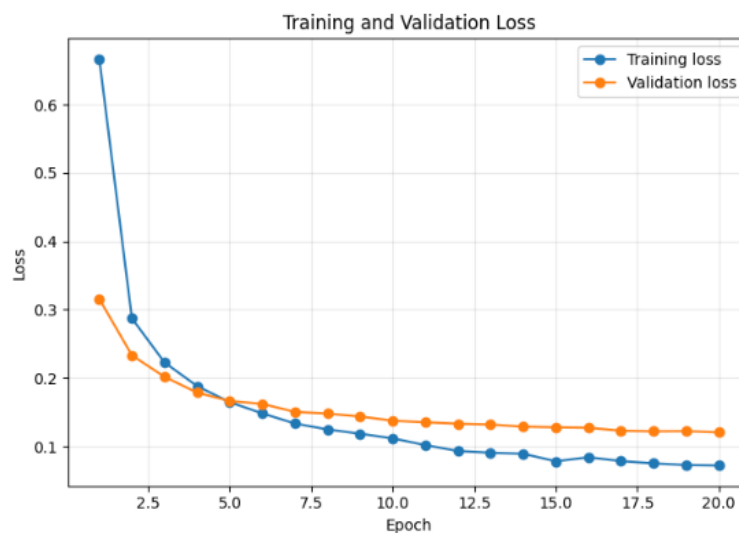
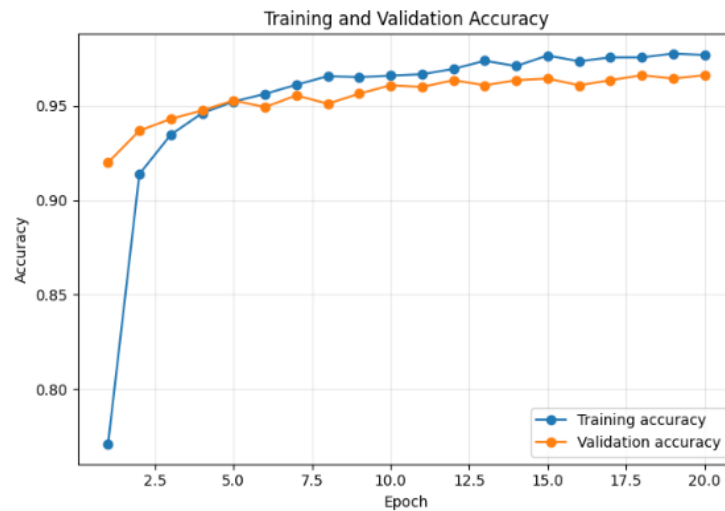
- Early stopping to monitor validation loss and restoring the weights from the best validation epoch.
- ReduceLROnPlateau to reduce the learning rate when validation loss stopped improving, allowing the optimiser to make smaller updates during periods of slower improvement.

The model completed 20 training epochs. At the final recorded epoch, it achieved:

Metric	Result
Training Loss	0.072
Training Accuracy	97.69%
Validation Loss	0.1205
Validation Accuracy	96.62%

The best validation loss was obtained during the final epoch, and the callback restored the weights from the last epoch. The difference between training and validation accuracy was relatively small, suggesting that the model did not suffer from severe overfitting. However, the training accuracy remained slightly higher than

the validation accuracy, which is expected because the model directly learned from the training images.



The training and validation curves should be interpreted together. A healthy training process is indicated by increasing accuracy and decreasing loss. A large and persistent gap between training and validation performance would indicate possible overfitting. In this experiment, the curves remained relatively close and validation performance improved throughout most of the training process.

The model was finally evaluated on an independent test set containing 562 images. It achieved an overall test accuracy of 95.73%.

Metric	Result
Test Accuracy	95.73%
Average Precision	94.29%

Average Recall	93.62%
Average F1-score	93.92%

Precision measures how many images assigned to a class were actually members of that class. Recall measures how many images belonging to a class were successfully detected. The F1-score is the harmonic mean of precision and recall.

Performance by class was as follows:

Class	Precision	Recall	F1-score	Support
Unripe	92.31%	88.89%	90.57%	27
Freshunripe	100%	100%	100%	83
Freshripe	92.52%	97.06%	94.74%	102
Ripe	85.71%	80.77%	83.17%	52
Overripe	97.37%	98.23%	97.8%	113
Rotten	97.81%	96.76%	97.28%	185

The best results were obtained for freshunripe, with an F1-score of 100%, and overripe and rotten, with an F1-score of approximately 97%. The weakest class was ripe, with an F1-score of approximately 83%. This suggests that intermediate ripeness stages are more difficult to distinguish from neighbouring categories than clearly unripe or rotten bananas.

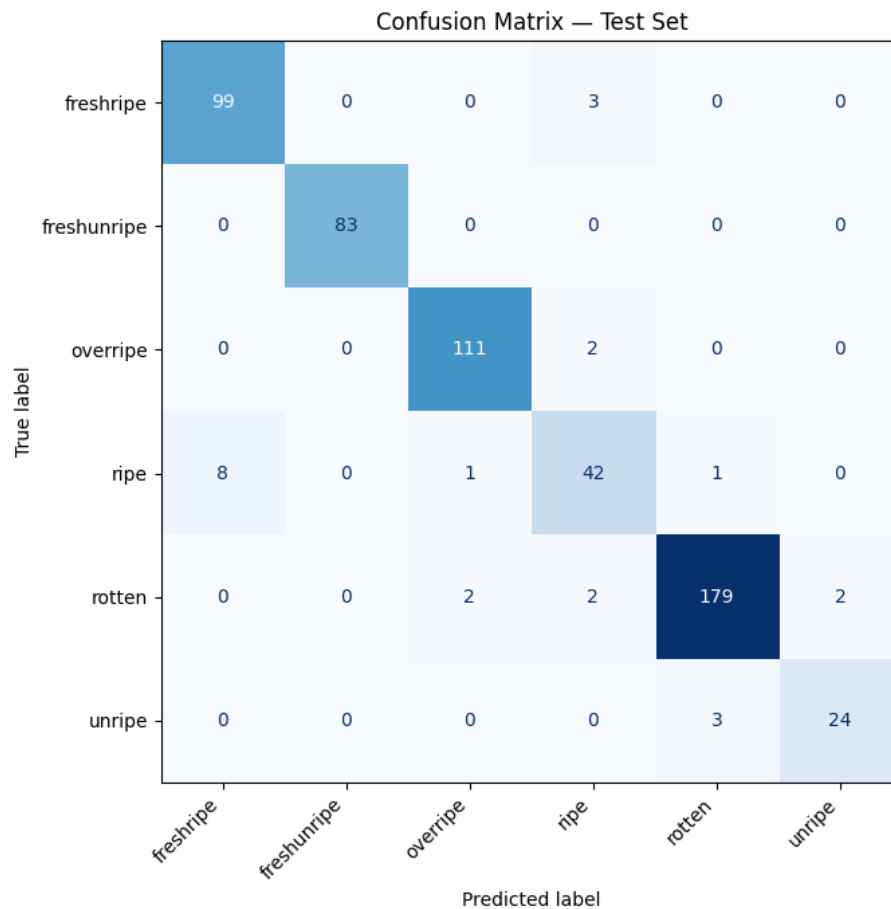
The unequal support across classes should also be considered. In particular, the unripe class contained only 27 test images, so a small number of errors have a relatively large effect on its recall and F1-score.

The confusion matrix showed that the model correctly classified 538 out of 562 test images. The most important errors involved visually similar ripeness stages. In particular, some ripe bananas were classified as freshripe, and some freshripe bananas were classified as ripe.

These errors are plausible because ripeness is a continuous biological process rather than a set of completely separated visual categories. Bananas close to the boundary between two categories may have similar peel colours and similar numbers of brown spots.

The model performed particularly well on rotten and overripe bananas, this is relevant to the smart-fridge application because detecting bananas that are likely

to be spoiled could support an early intervention, such as a reminder, recipe suggestion, or notification to consume the product soon.



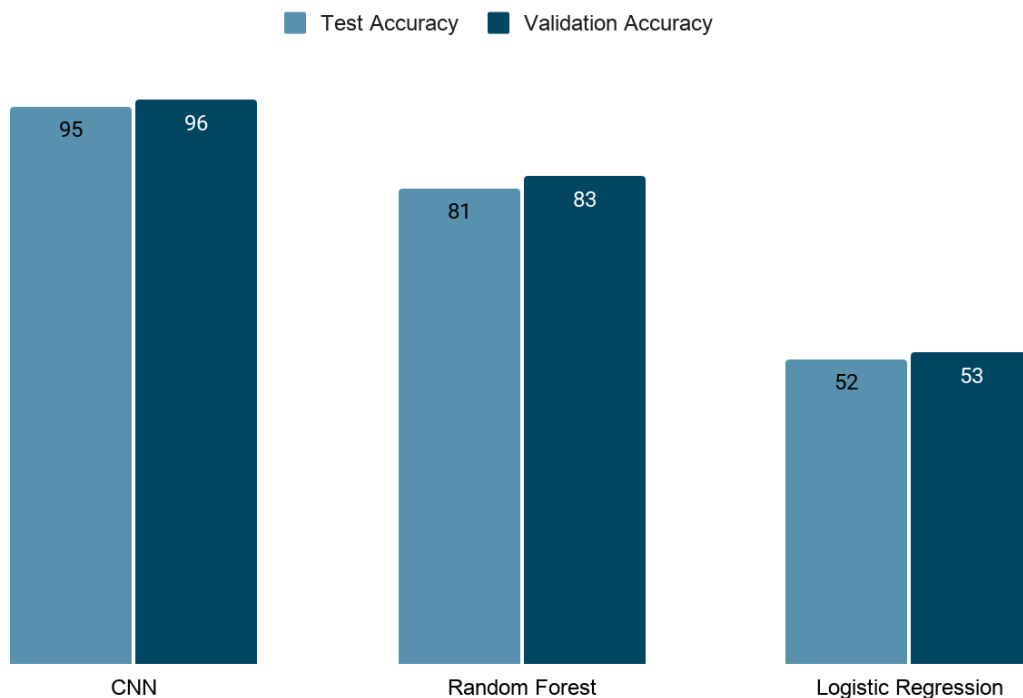
Part 6.6 - Model Comparison

The CNN model achieves the highest performance with a 95% Test Accuracy and 96% Validation Accuracy. It significantly outperforms Random Forest by +14.74% and Logistic Regression by +45.26% in test accuracy.

Random Forest achieves a 81% Test Accuracy and 83% Validation Accuracy. While it performs well, it stands behind CNN 8 (-14.74%), but after Logistic Regression (+55.77% relative improvement over Logistic Regression).

Logistic Regression shows the lowest accuracy with 52% Test Accuracy (-45.26% against CNN and -35.80% against Random Forest) and 53% Validation Accuracy. This performance suggests that a linear model is inadequate for capturing the underlying complexity or non-linear patterns in this dataset.

Test and Validation Accuracy



Pairwise Test Accuracy Matrix (Relative Change)

The matrix below shows the relative percentage variation (%) when comparing a Target Model (row) against a Baseline Model (column) for Test Accuracy.

Target Model / Baseline	Baseline: CNN 95%	Baseline: Random Forest 81%	Baseline Logistic Regression 52%
CNN 95%	-	+17.28%	+82.69
Random Forest 81%	-14.74%	-	+55.77%
Logistic Regression 52%	-45.26%	-35.80%	-

This model comparison suggests that deep learning techniques are more effective in capturing hidden patterns in an image classification task than machine learning algorithms. Despite the fact that the improvement in accuracy is relatively effective from Logistic Regression to Random Forest, with +55.77% in accuracy, the accuracy remains at a low/mid rate with 81%, meaning that the model is not consistent.

Moving from Random Forest to CNN, there is a variation of +17.28%, and the overall accuracy reaches 95% stating that the model can be considered quite consistent with respect to the previous results obtained.

Considering that these metrics have been achieved with minimal parameter configuration, it might be a future objective to test the same CNN architecture with a different configuration of parameters.

Part 7 - Prediction Examples

Part 7.1 - External Images

After training and test-set evaluation of the CNN, external banana images were uploaded to assess whether the classifier could generalise beyond the original dataset. These images were not part of the training, validation, or test folders.

For each external image, the following steps were applied:

1. The image was loaded and converted to RGB format.
2. It was resized to 224×224 pixels.
3. A batch dimension was added, producing an input shape of (1, 224, 224, 3).
4. The image was passed to the trained model.
5. The model returned six class probabilities.
6. The class with the highest probability was reported as the prediction.

The MobileNetV2 preprocessing was already included in the model, so the external image was not normalised a second time before prediction.

Part 7.2 - Classification Results

Five external banana images were used to test the CNN model beyond the original dataset.

The model correctly classified two images, resulting in an external accuracy of 40%. It correctly identified a dark banana as overripe with 89.29% confidence and a yellow/green banana as freshripe with 99.67% confidence. However, the model misclassified the green/yellow banana, the yellow banana, and the green banana. The yellow banana was classified as freshripe with a very high confidence of 99.90%, although its true class was ripe. This high-confidence error indicates that the softmax confidence should not be interpreted as a guarantee of correctness.

External image	Predicted class	Confidence	True class
Dark banana	overripe	89.29%	Overripe
Yellow/Green banana	freshripe	99.67%	Freshripe
Green/Yellow banana	freshripe	55.88%	Freshunripe
Yellow banana	freshripe	99.90%	Ripe
Green banana	freshripe	72.62%	Unripe

Example of classification of external image (Green/Yellow banana):

Prediction: freshripe
Confidence: 55.88%



	class	probability	percentage
0	freshripe	0.5588	55.88%
1	freshunripe	0.4212	42.12%
2	ripe	0.0198	1.98%
3	rotten	0.0001	0.01%
4	unripe	0.0001	0.01%
5	overripe	0.0000	0.00%

Part 7.3 - Misclassification Analysis

The errors mainly involved neighbouring ripeness categories, especially freshripe, ripe, freshunripe, and unripe. This is consistent with the confusion matrix obtained on the internal test set, where intermediate ripeness stages were also more difficult to distinguish. The external images may also differ from the training data in terms of lighting, background, image quality, banana variety, and photographic style. Therefore, the external test suggests that the model performs well on clearly distinguishable examples but requires further testing and possibly a more representative dataset before deployment in a smart-fridge application.

Part 7.4 - Model Generalisation

All three models display minimal degradation between Validation Accuracy and Test Accuracy (a gap of only 1% to 2%), indicating promising generalization ability without significant overfitting.

Part 8 - Business Application

Part 8.1 - Business Context

The problem

Household food waste constitutes a significant socio-economic and environmental issue, with residential kitchens accounting for a substantial proportion of overall disposals. Fresh product represents a particularly vulnerable category that is dominant in the kitchen bins and needs further attention. It is distinguished by its limited shelf-life and high sensitivity to environmental factors.

A key driver of domestic waste is the inability of consumers to constantly monitor and manage degradation stages of highly perishable fruits, such as bananas. Sometimes visual differences are unnoticeable, and inspection is hard to consumers or even disregarded, resulting in poor storage decisions or disposal of still edible fruit.

The user

The primary end-users of the system are residential consumers, defined as health-conscious, environment friendly and tech-savvy households seeking to optimize produce consumption, minimize kitchen food waste and automate meal planning with the use of modern equipment.

At an enterprise level, the target group is smart home appliance manufacturers and specifically refrigerator manufacturers with technology integration interest. The enterprises might be able to support the integration of the computer vision model into the next generation of fridges and other residential appliances to deliver differentiated, AI-driven solutions.

An extended user group consists of commercial food service operators, including restaurant kitchens, bars and catering services. In these high-volume commercial settings, integrating the classification engine into professional-grade appliances and cold storage hardware enables automated, real-time inventory tracking. This institutional integration reduces raw material degradation, assists culinary staff in first-in, first-out (FIFO) inventory rotation, and drastically decreases food prep waste.

Part 8.2 - Potential Use Cases

The proposed image recognition model is integrated as a hybrid computer vision service, deployed either directly at the edge within embedded appliance hardware (e.g., smart refrigerator interior cameras) or via a cloud REST API connected to a companion mobile application. In smart appliances, internal wide-angle, low-light cameras capture periodic snapshots of crisper compartments whenever doors are sealed. In mobile deployments, users capture product photos via an intuitive smartphone interface. The lightweight classification engine processes raw pixel data, maps fruit degradation states, and interfaces with a notification service to trigger automated, context-aware user recommendations.

1. Image Capture & Preprocessing: An internal appliance camera captures an image upon door closure (or the user captures a photograph via mobile app). The image undergoes automated cropping, lighting normalization, and resizing.

2. Model Feature Extraction & Inference: The normalized frame is passed into the trained model, running locally on edge hardware or processed asynchronously via cloud endpoints.

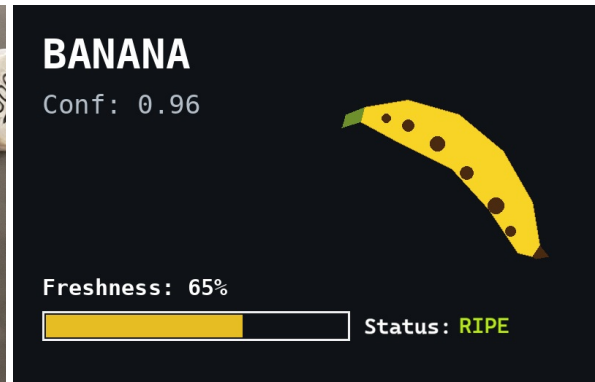
3. Class & Ripeness Prediction: The model predicts the item category and assigns a degradation class (e.g., *Banana* → *Class: Fresh-Ripe*) alongside a statistical confidence score.

4. Inventory & Expiry Logging: The application updates the local digital kitchen inventory, establishing an estimated remaining shelf-life window (e.g., 24–48 hours before decay).

5. Actionable Recommendation: The application triggers an automated push notification or display prompt recommending immediate consumption, presenting curated waste-reduction recipes (e.g., banana bread or smoothies), or advising culinary staff to prioritize the item for FIFO inventory rotation.



Sample monitor device in fridge door



Sample application view with banana reading

Part 8.3 - Business Benefits

The Value: The adoption of this computer vision system delivers multi-faceted economic, operational, and environmental value across both residential and commercial domains.

Cost Reduction & Waste Mitigation: Average households lose significant financial resources annually to unconsumed food waste. By converting subjective visual tracking into timely alerts and automated recipe pairing, the system reduces fresh produce spoilage by up to 30%, delivering direct household savings.

Automation of Inventory Management: The system eliminates manual inventory logging in domestic kitchens and commercial setups. Automated image-based tracking replaces subjective human checks, minimizing human error in spoilage estimation.

Decision Support & Behavioral Change: Rather than acting as a passive classifier, the application serves as an active decision-support tool to drive healthy consumption habits and timely utilization.

Commercial Operational Efficiency: In professional kitchen environments, automated freshness tracking assists culinary staff in enforcing First-In, First-Out (FIFO) food safety compliance, lowering raw material waste costs, and standardizing quality control across shifts.

Production Differentiation & Sustainable Value: For smart appliance manufacturers, embedding real-time fresh produce intelligence provides a compelling, premium software feature that elevates product value while directly advancing UN Sustainable Development Goal Target 12.3 (halving food waste).

Part 9 – Limitations and risks

Several limitations and risks should be considered before applying the system in a real smart-fridge or commercial environment. These limitations concern the input features, the dataset, the image-acquisition conditions, and the possible consequences of incorrect predictions.

Part 9.1 - Model and data limitations

Feature Extraction Constraints (Baseline ML): The logistic Regression (52% accuracy) and the random Forest (82% accuracy) rely on global mean RGB and HSV color features, the problem with those global color averages is that ones lose spatial context, causing the model to confuse localized rot or spot patterns with overall green or yellow skin hues. As a result, the models cannot determine where a colour occurs in the image. A small area of rot or a local cluster of brown spots may therefore be averaged together with the rest of the green or yellow peel. This can cause the baseline models to confuse local deterioration with the overall colour of the banana.

The CNN-based models are better suited to this task because convolutional filters can learn local patterns such as spots, edges, textures, and damaged areas.

Label Ambiguity: Banana ripening is a continuous biological process, whereas the dataset represents it using six discrete categories. This creates unavoidable borderline cases (such as ripe vs. overripe or overripe vs. rotten), that suffer from subjective human labeling and high overlap in color values, introducing label noise and limit the maximum achievable performance.

Severe Class Imbalance: The dataset is not perfectly balanced. The dataset heavily favors the rotten class (34.1% - 1,913 images), while the unripe class is critically underrepresented (3.4% - 191 images). This imbalance can make overall accuracy more influenced by the better-represented classes and can make performance estimates for smaller classes less stable.

Limited scope of the image setup: The dataset consists solely of single, centered bananas on simple backdrops. The current setup cannot detect or classify bananas in clusters, partially occluded fruits, or mixed fruit. A model trained mostly on isolated objects may perform less reliably when the smart-fridge camera captures a realistic shelf or drawer containing several products.

Possible Background Correlations: The images contain different backgrounds, such as wood, tiles, and white surfaces. This variation is useful, but it also creates a potential dataset-bias risk. If certain ripeness classes were photographed more frequently against particular backgrounds, a model could

learn correlations between the background and the label rather than relying only on the banana's appearance.

This risk was not explicitly tested in the current project. It could be investigated in future work using background randomisation, controlled image collection, or tests on images with unseen backgrounds.

Part 9.2 - System and deployment risks

Environmental and Lighting Sensitivity: The models heavily depend on color statistics (RGB/HSV). Changes in real-world ambient lighting, low-light fridge conditions, shadows, different camera color reproduction or condensation or dirt on camera lens will skew color features and lead to incorrect predictions.

Single-Angle Misclassification: Rot or bruising often starts on one side of a banana. A single-camera snapshot (e.g., a smart fridge door camera) viewing only the unblemished side will misclassify a rotting banana as fresh.

Downstream Business Risks (Smart Home and Commercial Use): Incorrect predictions may create two main types of business risk.

- *False positives* (rotten/overripe when the banana is still suitable for consumption): prematurely alerting users to discard or use the bananas that are still fresh. It leads to unnecessary food waste, defeating the project's primary goal.
- *False Negatives* (ripe/fresh when the banana is deteriorated): failing to flag decaying produce in commercial or smart fridge setups can result in bad food, unpleasant odor, and loss of user trust in automated inventory systems.

Part 9.3 Risk Mitigation Strategies

Use spatially aware deep-learning models: CNN-based models, including the transfer-learning MobileNetV2 model used in this project, can learn local spatial patterns such as spots, texture changes, edges, and damaged regions. They are therefore more appropriate for image classification than models based only on global colour averages.

Improve class balance: possible strategies include class-weighted loss functions, focal loss or targeted data augmentation (oversampling unripe images) to balance prediction performance.

Use confidence thresholds and human review: the system should retain all class probabilities rather than reporting only the most likely class. If the highest probability is below a predefined threshold, the system could request another

image, capture the banana from a different angle, ask the user to confirm the ripeness or display the top two or three predicted classes.

Part 10 - Conclusion

Part 10.1 - Future Improvements

To face possible overlapping classes during classification tasks, the project aimed to incrementally evaluate techniques in order to understand which one was best-suited for this task.

From modelling linear relationships to non-linear relationships, or manually extracting features and automatically extracting features, the goal was to assess different algorithms on a possible real case-scenario.

Nevertheless, Artificial Intelligence can dramatically improve this classification tasks, where connecting through an API a generative model, could not only give us the state of ripeness or decay of bananas, but also other interesting characteristics (like provenance from size and colours) and possible recipes for meal (thanks to the generative content provided by AI).

Part 10.2 - Conclusion

This project addresses the challenge of food waste and subjective manual inspection by developing an automated image-classification system to determine the visible ripeness stage of bananas. Recognizing that human vision is frequently compromised by fatigue, inconsistent standards, or variable lighting, the study utilizes a multi-class classification task to categorize fruit into six distinct states: unripe, freshunripe, freshripe, ripe, overripe and rotten. To accomplish this, the project leveraged a dataset comprising 5,616 labeled images split into training, validation, and test subsets. This dataset inherently captures the complexities of a continuous biological ripening process discretized into hard boundaries, alongside notable class imbalances, such as the disproportionate prevalence of rotten samples compared to underrepresented unripe ones.

To solve this task, an incremental modeling approach was adopted, benchmarking traditional machine learning models against deep learning architectures. Initial baseline models utilizing global RGB and HSV color statistics and specifically Multinomial Logistic Regression and Random Forest, achieved test accuracy of 52% and 81% respectively. However, because global color averages forfeit spatial context and struggle with localized browning, a convolutional neural network (CNN) utilizing a transfer-learning MobileNetV2 base was implemented. Freezing the pre-trained feature extractor and training a

custom classification head yielded exceptional performance, attaining a test accuracy of 95.73% alongside robust precision, recall, and F1-scores across most categories.

From a practical perspective, this technology offers significant commercial and residential utility, ranging from smart-fridge kitchen management and automated grocery inventory tracking to supply-chain quality control and recipe recommendation systems. Despite its high accuracy, the deployment of the model carries notable limitations and risks, including sensitivity to ambient lighting, single-angle viewing constraints, and potential misclassifications near continuous biological class boundaries. Consequently, future improvements should incorporate confidence thresholds, multi-angle imaging, and enhanced data balancing strategies to ensure reliable real-world performance without compromising user trust.

Appendix

Database choosen:

<https://universe.roboflow.com/roboflow-universe-projects/banana-ripeness-classification>

Google Colab Notebook:

https://drive.google.com/file/d/1EdA9n8_bowdmPuppNUoLiwfMIpKyz0pB/view?usp=drive_link

Image Gallery

<https://www.istockphoto.com/vector/kitchen-interior-smart-fridge-and-sink-on-tabletop-gm1223239497-359253655>

Images generated using prompts about banana ripeness monitor system by Gemini, 3.5 Flashlight, Year 2026, <https://gemini.google.com/>

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